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In [1]:

```
#Selection Sort
#34 45 2 12 6 9 67
#Assumption
#min = 34
#iterate the rest of the array to find the new minimum

#2 45 34 12 6 9 67
#min = 45
#iterate the rest of the array to find the new minimum
#min = 6
#2 6 34 12 45 9 67

#and so on
```

In [6]:

```
def selectionsort(array, size):
    print(array)
    for step in range(size):
        min_index = step
        for i in range(step + 1, size):
            if array[i] < array[min_index]:
                min_index = i
        temp = array[step]
        array[step] = array[min_index]
        array[min_index] = temp
        print(array)


array = [34, 45, 2, 12, 6, 9, 67]
selectionsort(array,len(array))
print('Selection sort, Sorted array is as follows::')
print(array)
```

```
[34, 45, 2, 12, 6, 9, 67]
[2, 45, 34, 12, 6, 9, 67]
[2, 6, 34, 12, 45, 9, 67]
[2, 6, 9, 12, 45, 34, 67]
[2, 6, 9, 12, 45, 34, 67]
[2, 6, 9, 12, 34, 45, 67]
[2, 6, 9, 12, 34, 45, 67]
[2, 6, 9, 12, 34, 45, 67]
Selection sort, Sorted array is as follows::
[2, 6, 9, 12, 34, 45, 67]
```

In [27]:

```
#Heap Sort      Time Complexity --> O(nlogn)
#Max heap
#Min heap
def heapify(array, n, i):
    largest = i
    lc = 2*i + 1
    rc = 2*i + 2

    if lc < n and array[i] < array[lc]:
        largest = lc
    if rc < n and array[largest] < array[rc]:
        largest = rc
    if largest != i:
        array[i], array[largest] = array[largest], array[i]
        heapify(array, n, largest)


def heapsort(array):
    n = len(array)

    #Max heap
    for i in range(n//2,-1,-1):
        heapify(array,n,i)


    for i in range(n-1,0,-1):
        array[i], array[0] = array[0], array[i]
        heapify(array,i,0)


array = [34, 45, 2, 12, 6, 9, 67]
heapsort(array)
print('Heap sort, Sorted array is as follows::')
print(array)
```

```
Heap sort, Sorted array is as follows::
[2, 6, 9, 12, 34, 45, 67]
```

In [ ]: